

Functional specification

Priority v2 skills: compile shell + install shell

Version	0.1
Date	2026-09-18
Status	Ready to implement behind WP0 pins
Scope	v2 only. Do not change repo-root v1 (src\Prepare-NamedForm.ps1).
Catalog	https://mcp-priority.nts.a.uk (grab-only; no WCF/SQL)
Executor	WCF preferred. WINRUN only if WP0 proves file step cannot be fed.
Slogan	Red before WCF.

Grok bot does not run on production servers. Execution is local against user-allowlisted instances.

1. Purpose

Add two executable skills next to `priority-formprep` so an agent can **compile** a Version Revision on an allowlisted instance into `NN.sh`, and **install** a caller-supplied shell from a **build set** onto an allowlisted instance. Catalog MCP stays grab-only.

2. Actors and machines

User	Fills instance allowlist; supplies revision id or build-set path; sets CredMan
Catalog MCP	Serves SKILL.md, schemas, runner files. No WCF, no SQL
Local plugin / runner	list_instances, compile_shell, install_shell, prepare_form
Proof instance	CE DEV (or equivalent sandbox). Only place fixtures may exist
Target instance	Any allowlisted row with allowLive respected

{machine} = instance_id from the allowlist. Never invent webBaseUrl or SQL. Default allowlist: %USERPROFILE%\priority-formprep\instances.json (override PRIORITY_FORMPREP_INSTANCES).

3. Skills and tools

catalog/priority-formprep	prepare_form	FORMPREPDRCT2 via EFORM	instance_id, name
catalog/priority-shell-compile	compile_shell	Prepare Upgrade [ENAME: WP0]	instance_id, revision
catalog/priority-shell-install	install_shell	Install Upgrade [ENAME: WP0]	instance_id, shell, optional allow_dbi

One plugin may host all three tools. Catalog gets two new folders so list_catalog shows compile and install separately. Do not implement Prepare Upgrade and Install Upgrade as one tool.

4. Functional requirements

4.1 Common

FR-C1. Refuse unless instance_id exists in the allowlist (reason=instance_unknown, exit 2).

FR-C2. Refuse live-looking instance when allowLive is false (reason=live_refused).

FR-C3. CredMan only. Missing cred → reason=no_cred, no WCF.

FR-C4. Preferred executor: WCF / Web SDK walk. WINRUN only if WP0 proves the file step cannot be fed over WCF.

FR-C5. SDK/procedure “successfully completed” is not success.

FR-C6. Result JSON on stdout (last {...} line parseable). No passwords in JSON or logs.

FR-C7. One unit of work per call (one revision, one shell, one form).

FR-C8. Windows runner. Non-Windows → reason=windows_only.

FR-C9. Do not modify v1 paths or v1 success gate.

4.2 compile_shell

FR-P1. Input: instance_id, revision (integer or string NN as used on Version Revisions).

FR-P2. Resolve revision on that instance. Missing → reason=revision_missing, errors[] set, no file claimed.

FR-P3. Run Prepare Upgrade for that revision over WCF.

FR-P4. ok=true only if the shell file exists (instance or runner-controlled build-set path) and size > 0.

FR-P5. Return path suitable to pass straight into install_shell.shell.

FR-P6. On fail, errors[] never empty (sdk messages and/or “shell not created”).

FR-P7. Do not create revisions or entities. Compile only existing revisions.

FR-P8. Do not install.

4.3 install_shell

FR-I1. Input: instance_id, shell (required). shell is a path on the runner / build set, not “latest file in system\upgrades”.

FR-I2. Optional allow_dbi default false.

FR-I3. Parse the file before WCF. Not a Priority shell → reason=parse_failed, no install.

FR-I4. Path allowlist: build-set roots and instance agentWork. Refuse .., foreign UNC, catalog fixtures/ live install.

FR-I5. Stage the file onto the target instance allowlisted dir, then feed that server path to Install Upgrade over WCF.

FR-I6. If parse shows DBI and allow_dbi is false → reason=dbi_refused, no WCF.

FR-I7. ok=true only if the SQL gate on that instance passes (§6).

FR-I8. On any fail, errors[] never empty.

FR-I9. Do not call FormPrep. If the shell names forms, return postInstall.formsUnprepared[] for prepare_form.

FR-I10. Do not create fixture entities.

5. Tool schemas

```
compile_shell { instance_id: string, revision: string }
install_shell { instance_id: string, shell: string, allow_dbi: false }
```

6. Success gates

compile_shell — ok=true iff procedure reached end without Blocker messages, output .sh exists and length > 0, and path is returned.

install_shell — ok=true iff all of: procedure reached end without Blocker messages; **[WP0]** install-log / version-revision row for NN advanced after startedAt (or equivalent pinned table); every TAKESINGLEENT name parsed from the shell exists in T\$EXEC on that instance; no errors[].severity=Blocker.

Do not invent table names in product code. WP0 writes pin.psd1 / skill config.

7. Result JSON

Common (align with FormPrep v2): ok, runId, instanceId, skill, executor, startedAt, endedAt, durationSec, reason, exitCode, errors[], whatIf?

errors[]: source, severity (Info|Warning|Blocker), text, optional entityHint. On fail, errors.length >= 1. On success, omit empty error-form scrapes.

Compile extras: revision, path, bytes. Install extras: path (source), stagedPath, revision, codes[], dbi, installed, gate, postInstall.formsUnprepared[].

Exit: 0 ok or WhatIf; 2 refuse (no mutation); 3 ran, gate or proc failed.

FormPrep only	prepared
Compile	compiled, revision_missing, shell_not_created
Install	installed, parse_failed, path_refused, dbi_refused, already_installed (optional), proc_failed, gate_unchanged, partial_entities
Shared	instance_unknown, live_refused, no_cred, windows_only, whatIf, pin_incomplete

8. Errors (install, required)

Populate errors[] from, in order, whatever WP0 finds exists:

1. WCF / SDK procedure messages (source=sdk)
2. Upgrade/revision error form or report if the proc opens one
3. Gate failures (source=gate) — e.g. entity in shell missing from T\$EXEC
4. Parse / path / DBI refuses (source=parse | policy)

Never return ok=false with errors=[].

9. Build-set parameter

The shell is an artefact of the build set. The caller passes it. Compile instance and install instance may differ. The file moves; the skill does not assume a shared system\upgrades. MVP: one .sh per install_shell. No directory install.

```
compile_shell({ instance_id: "ce-dev", revision: "1042" })
  → path \\share\buildset\1042.sh
```

```
install_shell({ instance_id: "ce-dev", shell: "\\share\buildset\1042.sh" })
  → ok + errors[]
```

10. Hard rules (SKILL.md)

1. Never report compiled/installed unless the gate in §6 holds.
2. Never SQL-fake an install or compile.
3. Never invent instance URLs.
4. Never log passwords.
5. Never install fixtures from git onto a non-proof instance.
6. Never create ZCLA_AGENT_SI_* (or any fixture) from these tools.
7. Never auto-prepare forms after install.
8. Never run when allowLive is false and the row looks live/PRI.

11. Repo layout (v2)

```
v2/apps/mcp-catalog/catalog/priority-shell-compile/  
  meta.json, SKILL.md, instance-schema.json  
v2/apps/mcp-catalog/catalog/priority-shell-install/  
  meta.json, SKILL.md  
v2/plugins/priority-formprep/          # extend tools OR add sibling plugin  
  scripts/Compile-Shell.ps1  
  scripts/Install-Shell.ps1  
  scripts/lib/                        # allowlist, sql, cred, parse-sh  
  fixtures/                          # sanitized .sh for parser tests only  
.grok-plugin/marketplace.json
```

Adding a catalog folder must not require MCP handler edits (list_catalog already scans meta.json).

12. Out of scope (MVP)

Creating Version Revisions / flagging steps; INSTITLE / language dictionaries; product Upgrade System from Help; headed Playwright; multi-file folder install; live DBI install (flag exists; proof later); changing v1; agentic_build integration.

13. WP0 pins

On the proof instance: read-only recon, then one human hand-compile and hand-install.

ENAME + type of Prepare Upgrade	compile WCF
ENAME + type of Install Upgrade	install WCF
Input step field for revision / file path	walker
Whether WCF can accept a server path	WCF vs WINRUN fallback
Install-log / revision table + columns	SQL gate
How DBI appears inside .sh text	parser
Error form/report opened by install	errors[]

Until pins exist, implement parser, allowlist, WhatIf, schemas, refuse paths, SKILL.md, and WP0 fail-fast tests. Do not call a guessed procedure name.

14. WP0 fail-fast tests

The pack must go red in seconds if pins are missing, guessed, or the proof instance is not ready. No WCF install/compile in this phase except a read-only recon query when an allowlisted proof instance is present.

Runner: v2/tools/Test-WP0.ps1 (first slice of Test-Pack.ps1). Exit 0 only if every gate passes. Exit 2 = pack not ready (do not start WP3/WP4). Exit 3 = recon ran and found a contradiction.

14.1 Always (no instance, CI and local)

WP0-T1 catalog	catalog/priority-shell-compile or catalog/priority-shell-install missing meta.json or SKILL.md
WP0-T2 schema	compile/install result schemas missing or invalid JSON Schema
WP0-T3 pin file	pin.json / pin.psd1 missing
WP0-T4 pin complete	PinComplete is true while any required pin is empty, UNKNOWN, or a guessed token without recon evidence
WP0-T5 pin incomplete	PinComplete is false AND runners would take the WCF success path (must refuse reason=pin_incomplete)

WP0-T6 no guessed ENAME	product scripts contain a hardcoded procedure name not read from pin
WP0-T7 parse fixture	sanitized fixture .sh missing or parser does not return revision + codes
WP0-T8 refuse	unknown instance, live refuse, missing shell, DBI without flag, path .. — exit 2 and zero WCF/winrun starts
WP0-T9 v1 frozen	repo-root src\Prepare-NamedForm.ps1 missing or WP0 patched it

If PinComplete=false, T5+T8 are the contract: tooling is skeleton-safe. That is a pass for early WP0, not a skip.

14.2 Proof instance present (allowLive=false)

Read-only SQL + optional WCF login probe. No Prepare Upgrade, no Install Upgrade.

WP0-R1 allowlist	proof instance_id not in instances.json or missing CredMan
WP0-R2 sql	cannot SELECT TOP 1 from T\$EXEC on that instance
WP0-R3 titles	zero T\$EXEC candidates for Prepare Upgrade / Install Upgrade / Version Revisions
WP0-R4 pin match	PinComplete=true but pinned ENAME not in T\$EXEC on this instance
WP0-R5 upgrades dir	configured upgrades / agentWork path not reachable from the runner
WP0-R6 wcf probe	login to webBaseUrl fails — stop, do not invent another URL
WP0-R7 contradiction	pin says WCF file-step works but walk transcript has no file/path input (or the reverse)

R3 does not require the human dummy entity. It only requires dictionary titles for the standard programs.

14.3 Fail-fast in the runners

```
if (-not $pin.PinComplete) { return reason=pin_incomplete; exit 2 }
if ($pin.PrepareUpgradeEname -in $null, '', 'UNKNOWN') { return reason=pin_incomplete; exit 2 }
# same for Install Upgrade ENAME and gate SQL id
```

A build agent that hardcodes a name to get past WP0 fails T4/T6.

14.4 Evidence

WP0 writes (no secrets): tests/wp0-last.json (gate id, pass/fail, candidate ENAMES) and docs/wp0-recon.md (only when R3 found rows). PinComplete flips to true only by a human after reading recon. Tests fail if a commit sets it true without pin fields populated.

14.5 Order in Test-Pack

1. WP0-T* (always)
2. Off-instance parser / schema / refuse (§15)
3. WP0-R* if proof instance env is set (PRIORITY_WP0_INSTANCE)
4. Stop. Do not run live compile/install ATs until WP0 is green and PinComplete=true.

Slogan for the build agent: red before WCF.

15. Acceptance

Off-instance

- Schema validates.
- Missing allowlist / unknown id / live refuse / bad path / DBI without flag → exit 2, no WCF.
- Fixture .sh parses to expected codes.

- WhatIf does not call WCF.

Proof instance (CE DEV)

- Human dummy revision compiled by tool → file exists.
- That file passed as shell installs → ok=true, gate moved.
- Truncated file → ok=false, errors[] non-empty.
- FormPrep prepare_form still works on an existing ENAME (no v1 change).

16. Decision log

1	Home: v2 catalog + local plugin. v1 frozen.
2	Fixture entity: proof instance only; not a product object.
3	Executor: WCF preferred; WINRUN only if file step cannot be fed.
4	DBI: parse-refuse unless allow_dbi. No idle-session MVP.
5	Shared kernel: v2 scripts/lib (allowlist, cred, sql). Do not extract v1.